

3.3 Part 2: Online Surveys with Avid Readers

In Part 2, we conducted an online study to understand avid readers' perceptions of authenticity, AI writing assistance, and work co-written by human writers and AI. We used writing work produced by six writers in Part 1, each of which represents a unique literature genre (P8: science fiction/fantasy, P10: biography, P12: poetry, P14: spiritual, P15: romance, and P19: screenplay), as materials for Part 2.¹ These include anonymized writing samples that writers submitted in the pre-interview survey of Part 1 (i.e., work written by themselves) as well as the two passages they wrote with the personalized and non-personalized AI during Part 1.

Participants of Part 2 (i.e., readers) first responded to a few open-ended questions about their attitudes toward AI-assisted writing. They then read three writing passages from the same writer (their solo work, work with personalized AI, and work with non-personalized AI) in randomized order. After reaching each article, participants assessed the writing with respect to *likeability* (how much they liked the writing), *enjoyment* (how much they enjoyed reading the writing), and *creativity* (how creative they thought the writing was). We chose these evaluation criteria as they were also used in prior work to assess AI-assisted writing (e.g., [32, 45]).

Next, readers performed two sets of comparisons with the three pieces of writing. First, they were informed that *some* of these pieces were written with AI, and were asked to compare and rate how likely each piece was to have been written by a human writer alone, or with the help of an AI writing support tool. Second, readers were informed which piece was the writer's solo work. They then compared the two AI-assisted writings by responding to the question: *"Compared to the text written independently by the author, to what extent do you think the co-written text preserves the authentic voice of the author?"* They were also asked to identify in which part(s) of the writing the author might have adopted AI assistance. Readers were blinded to the two different AI conditions; namely, they did not know which passage was co-written with a personalized versus a non-personalized AI tool, or that personalization was the variable being studied. We applied this blinding approach to ensure participants' assessments were based on the content per se and were not affected by their perceptions of personalized AI assistance.

After rating and comparing writing passages, participants responded to a few questions about their general opinions about AI-assisted writing. Table 3 presents the procedure of Part 2 and the variables measured throughout. Additionally, the full Part 2 questionnaire is attached in Appendix G and descriptive statistics of numeric measures are reported in Table 4. The entire Part 2 study was conducted online and took around 45 - 60 minutes to complete. Participants received cash compensation for their participation. Once again, the study protocol was reviewed and approved by the IRB of the authors' affiliation.

3.3.1 Recruitment and participants. We recruited participants on Reddit to find avid readers who regularly read the particular literature genres written by the six writers chosen from Part 1. One of the researchers joined multiple subreddits dedicated to discussions about reading.² (See Appendix E for a list of these subreddits.) Recruitment messages were posted on discussion threads in these subreddits. Interested individuals first filled out an interest form that surveyed the literature genres they were interested in and how frequently they read. They also reported the subreddit where they saw the recruitment message. 30.27% of those who filled out our interest form read on a daily basis, and 60.54% of them read weekly. We recruited participants who read their genres of interest at least once a week (Appendix F shows a list of genres that our reader participants regularly read. Note that each participant might frequently read more than one genre.). As prior research on Reddit has identified a large number of machine-generated responses, we showed

¹At the end of each session in Part 1, we informed writers and obtained their consent to use their writing for subsequent research.

²We received approvals from community moderators to post and participate in the discussions on these subreddits.

Step 1.) At the beginning of the study, participants responded to a few questions to express their opinions about AI writing assistance. These are single measures rated with 5-point Likert scales, including: • <i>Familiarity with generative AI:</i> Asking participants what they know about generative AI. • <i>Interest in reading AI-assisted writing:</i> Asking participants whether they are interested in reading work co-written with AI. • <i>Authenticity of AI-assisted writing:</i> Asking participants whether they consider AI-assisted writing as a writer's authentic work. Step 2.) Participants read three pieces of work from the same writer, including their solo work, work co-written with personalized AI, and work co-written with non-personalized AI. Participants were blinded from the three conditions. That is, they did not know which piece was done by the writers independently and which with AI assistance. Step 3.) Participants responded to a few questions to express their perceptions toward the three pieces of work while blinded from the three conditions. That is, participants did not know which piece was done by the writers independently and which with AI assistance when they responded to these questions. These are repeated measures; i.e., the same scales are used to rate each of the three pieces of work. • <i>Likeability:</i> Asking participants how much they like a piece of writing. • <i>Enjoyment:</i> Asking participants how much they enjoy a piece of writing. • <i>Creativity:</i> Asking participants how creative a piece of writing is. Step 4.) Participants were told some of these writings could have been done with AI assistance. They responded to a few questions to gauge which piece might be the writer's solo work and which piece might be co-written with AI. These are repeated measures; i.e., the same scales are used to rate each of the three pieces of work. • <i>Likelihood of human writing:</i> Asking participants to gauge whether a piece of writing was written by a writer independently or was co-written with AI assistance. Step 5.) Participants were told which piece of work was the writer's solo work and which two were co-written with AI, though we did not reveal which was done with personalized AI and which was done with non-personalized AI. Participants responded to a few questions to compare the two AI-assisted pieces to the writers' solo work. These are repeated measures; i.e., the same scales are used to rate the two pieces of AI-assisted work. • <i>Preserving writer's authentic voices:</i> Asking participants whether a piece of AI-assisted writing preserves a writer's authentic voice. • <i>Credits and authorship of work:</i> Asking participants how they might attribute the credits and authorship of a piece of AI-assisted work. Step 6.) Participants responded to a few questions to reflect on their opinions toward AI-assisted writing after reading some samples. These are single measures rated with 5-point Likert scales, including: • <i>Perception of the writing:</i> Asking participants' overall perception toward AI-assisted writing. • <i>Perception of the human writer:</i> Asking participants' perception toward writers who apply AI assistance to complete their work. • <i>Appreciation and evaluation of writing:</i> Asking participants whether they would appreciate a piece of AI-assisted work through different approaches.
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Table 3. The procedure and variables measured throughout Part 2 of the study.

Variable	Round 1	Round 1 (removed 7 possibly AI-generated responses)	Round 2
Opinions about AI writing assistance			
Familiarity with generative AI	3.85 ± 0.92	3.78 ± 0.96	3.65 ± 1.00
Interest in reading AI-assisted writing	3.58 ± 0.89	3.76 ± 0.91	3.69 ± 0.92
Authenticity of AI-assisted writing	3.70 ± 0.72	3.30 ± 0.98	3.44 ± 0.95
Reading writing samples			
Likeability			
Writer's solo work	3.80 ± 0.77	3.85 ± 0.77	3.88 ± 0.77
Work co-written with personalized AI	3.55 ± 0.51	3.7 ± 0.54	3.65 ± 0.54
Work co-written with non-personalized AI	3.75 ± 0.72	3.85 ± 0.72	3.74 ± 0.79
Enjoyment			
Writer's solo work	3.55 ± 0.89	3.63 ± 0.88	3.71 ± 0.91
Work co-written with personalized AI	3.40 ± 0.88	3.52 ± 0.89	3.53 ± 0.86
Work co-written with non-personalized AI	3.4 ± 0.88	3.59 ± 0.97	3.59 ± 0.96
Creativity			
Writer's solo work	3.70 ± 0.98	3.70 ± 0.99	3.76 ± 0.99
Work co-written with personalized AI	3.70 ± 0.80	3.59 ± 0.75	3.59 ± 0.70
Work co-written with non-personalized AI	3.25 ± 0.72	3.48 ± 0.80	3.47 ± 0.79
Comparing all writing samples			
Likelihood of human writing			
Writer's solo work	3.63 ± 1.01	3.73 ± 0.92	3.73 ± 0.91
Work co-written with personalized AI	2.84 ± 1.01	2.96 ± 1.00	3.00 ± 1.00
Work co-written with non-personalized AI	3.32 ± 1.2	3.19 ± 1.13	3.09 ± 1.10
Comparing personalized vs. non-personalized writing samples			
Preserving writers' authentic voices			
Work co-written with personalized AI	3.20 ± 0.68	3.32 ± 0.89	3.30 ± 0.82
Work co-written with non-personalized AI	3.6 ± 0.91	3.36 ± 0.90	3.37 ± 0.84
Credits and authorship of work			
Work co-written with personalized AI	2.90 ± 0.79	3.07 ± 0.83	3.09 ± 0.79
Work co-written with non-personalized AI	3.45 ± 1.00	3.30 ± 0.99	3.27 ± 0.94
Opinions after reading AI-assisted writing			
Perception of the writing	3.35 ± 0.97	3.30 ± 0.98	3.35 ± 0.91
Perception of the human writer	2.95 ± 1.25	3.48 ± 1.11	2.97 ± 1.25
Appreciation and evaluation of writing	3.50 ± 1.21	2.96 ± 1.21	3.56 ± 1.04

Table 4. Descriptive statistics of numeric variables in Part 2. Numeric values in all cells represent *Means ± S.D.*

some question texts in image format in the interest form to filter out bots that can only read text. We also included an open-ended question (i.e., "In one to two sentences, describe your favorite literature genre and why you favor it."). Two researchers reviewed these open-text responses and screened out those that were likely generated by machines, as detailed below.

3.3.2 Screening AI-generated responses. We first set out to collect feedback from three readers for each writer's work, resulting in $6 \times 3 = 18$ participants. However, we noticed a substantial number of open-text responses that might be AI-generated (e.g., by ChatGPT) after examining our initial batch of data. We discussed among our research team to identify the following criteria for screening: (1) Responses that were consistently written in a bullet-point style following the structure of "[Topic]: [Description]," which is a common style in ChatGPT output; (2) We used our survey questions as prompts and collected responses from popular generative AI tools, including ChatGPT, Microsoft Bing Chat, and Google Bard. We cross-checked whether participants' responses overlapped significantly with those AI-generated ones.